NATIONAL UNIVERSITY OF SINGAPORE
FACULTY OF SCIENCE

Experiment-Based Risk Assessment Form

Name of Department	**Department of Chemistry**	**Name and Location of Lab**	**S5-04**
Name of Laboratory	**Chemical Synthesis Teaching Lab**	**Name of PI (lecturer-in-charge)**	**Dr. Edith S. H. Chan**
Name of Student	**Choi Seung Eun**	**Name of Activity/Experiment**	**CM3191 Inorganic - Metallocene: Session II - Purification and Characterization of Ferrocene**

Hazard Identification				Risk Evaluation & Control						
No	**Description / Details of Steps in Activity**	**Hazard(s)**	**Possible Accident(s)**	**Existing Risk Control (Mitigation)**	**Severity**	**Likelihood (probability)**	**Risk Level**	**Additional Risk Control**	**Person Responsible**	**By (Date)**
1	Clean and dry all the glassware (beaker and petri dish base and cover) used for the purificaiton	**Physical hazard:** Broken glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps containiner. Handle and use glassware with care	2	1	2			
2	Purification of ferrocene. Record the weight of the glass petri dish cover with an analytical balance. Spread the crude ferrocene on a glass petri dish base and cover it with the glass petri dish cover. Place the petri dish on the hot plate.	**Physical hazard:** Broken petri dish	Cuts from broken petri dish	Visual inspection for any cracks before usage. Proper disposal of broken petri dish in Sharps containiner. Handle and use the petri dish with care.	2	1	2			
		Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
3	To purify the crude ferrocene, slowly raise the hot plate temperature (< 150 °C) until the ferrocene sublimes to the dish cover. Cool the dish cover by placing a 400 mL beaker filled with ice water on top of it to facilitate the sublimation.	**Physical hazard:** Placing 400 mL beaker on top of the petri dish cover	Spillage of ice water on the hot plate can cause rapid evaporation and skin burn	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 2. Carefully place the 400 mL filled with ice and water on the petri dish	2	1	2			
		Electrical hazard: Usage of hot plate and exposed wire of the cable	Electric shock if used improperly	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 2. Visually inspect apparatus for cracks before use 3. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction	2	1	2			
		Heat hazard: Usage of hot plate for sublimation	Contact with hot plate can cause skin burn	Turn on the hot plate after completing the set up of placing the petri dish and beaker on the hot plate. Turn off the hot plate after finishing ferrocene purification.	2	1	2			
4	Once the petri dish cover is filled with sublimed ferrocene, remove and place it so that the side containing ferrocene is facing up. Quickly cover the petri dish base with a new cover to prevent the loss of ferrocene. Allow the petri dish cover filled with sublimed ferrocene to cool down to room temperature	**Physical hazard:** Broken petri dish	Cuts from broken petri dish	Visual inspection for any cracks before usage. Proper disposal of broken petri dish in Sharps containiner. Handle and use the petri dish with care.	2	1	2			
		Heat hazard: Removing the heated petri dish cover from the hot plate	When removing the petri dish cover, can get skin burn from touching the heated petri dish cover or the hot plate	Slightly touch the petri dish cover to check whether it is not too hot. Carefully remove the petri dish cover while ensuring that fingers do not touch the hot plate.	2	1	2			

		Physical hazard / Human factor: Spillage of ferrocene	When flipping the petri dish cover so that the side with ferrocene is facing up, ferrocene can drop	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Clean the working area after handling 4. Thoroughly wash skin after handling	2	1	2			
		Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process. Ferrocene deposition on hands and arms while removing the petri dish cover	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
5	Record the weight of the petri dish cover filled with sublimed ferrocene. Transfer the sublimed ferrocene from the petri dish cover into a clean sample vial	**Physical hazard:** Broken petri dish	Cuts from broken petri dish	Visual inspection for any cracks before usage. Proper disposal of broken petri dish in Sharps containiner. Handle and use the petri dish with care.	2	1	2			
		Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Physical hazard / Human factor: Spillage of chemicals (ferrocene)	Spillage of chemicals when transferring it into a sample vial.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Handle the chemicals with care	2	1	2			
6	Characterisation of the purified ferrocene. Record the microscopic image of the purified ferrocene by using the stereomicroscope with the lowest magnification ratio of 14x.	**Chemical and fire hazards:** Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Physical hazard / Human factor: Spillage of ferrocene	Spillage of ferrocene during the recording of microscopic image	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Handle ferrocene with care	2	1	2			
		Electrical hazard: Exposed wire of the cable and/or broken plug or socket for the stereomicroscope	Electric shock if used improperly	1. Check the cable, plug and socket before use 2. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction 3. Refer to the instrumental operation manual	2	1	2			

7	Use agate mortar and pestle to grind 0.09-0.12 g of the purified ferrocene into fine powder, and use the magnetic susceptibility balance to measure both volume and mass susceptibilities of the purified ferrocene.	**Chemical and fire hazards:** Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process and the measurement of magnetic susceptability.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Physical hazard: Mortar and pestle breakage	Cuts and injury from broken mortar or pestle.	1. Visually inspect the mortar and pestle for any breakage before use. 2. Do not use excessive force when grinding ferrocene with mortar and pestle to prevent breakage 3. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment	2	1	2			
		Electrical hazard: Exposed wire of the cable and/or broken plug or socket for the magnetic susceptability balance	Electric shock if used improperly	1. Check the cable, plug and socket before use 2. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction 3. Refer to the instrumental operation manual	2	1	2			
8	Prepare the standard solution of ferrocene by weighing 35-40 mg of ferrocene with the analytical balance and dissolving it in methanol in a 25 mL volumentric flask. Record the UV-Vis spectrum (300 - 700 nm). Collect the spectra showing weak and strong peaks (A = 0.1 - 1.0) by diluting the standard solutions in the fume cupboard.	**Chemical and fire hazards:** Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Physical hazard: Broken volumetric flask	Cuts and injury from broken volumetric flask	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps containiner. Handle and use glassware with care	2	1	2			
		Chemical and fire hazards: Methanol is harmful, toxic, and highly flammable	Methanol is toxic when swallowed, inhaled or in contact with skin. Methanol is highly flammable.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard methanol into proper waste bottle 4. Keep methanol away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Electrical hazard: Exposed wire of the cable and/or broken plug or socket for the spectrophotometer and analytical balance	Elecric shock if used improperly	1. Check the cable, plug and socket before use 2. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction 3. Refer to the instrumental operation manual	2	1	2			

9	Obtain the solid IR spectrum (500 - 4000 cm-1) of the purified ferrocene by either ATR-IR or KBr disc.	**Chemical and fire hazards:** Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
		Chemical hazard: KBr is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment	2	1	2			
		Electrical hazard: Exposed wire of the cable and/or broken plug or socket for the spectrometer	Elecric shock if used improperly	1. Check the cable, plug and socket before use 2. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction 3. Refer to the instrumental operation manual	2	1	2			
		Radiation hazard: IR radiation	Injury from IR beam	Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment	2	1	2			
10	Transfer the purified ferrocene (~ 1 mg) into the NMR tube containing 0.6 mL of CDCl3.	**Physical hazard:** NMR tube	Injury from broken NMR tube	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Visually inspect the NMR tube for cracks before use 4. Handle the NMR tube with care and do not use too much force when capping the NMR tube	2	1	2			
		Chemical hazard: Deuterated chloroform (CDCl3)	Deuterated chloroform is carcinogenic and an irritant to the skin and eyes	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment	2	1	2			
		Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			
11	Transfer the purified ferrocene (1 - 2 mg) into a clean sample tube and submit the labelled tube for ESI-MS analysis.	**Chemical and fire hazards:** Ferrocene is harmful, toxic, irritating and flammable	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Discard ferrocene into proper waste bottle 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction. 6. Wash skin thoroughly after handling	2	1	2			

12	Washing of glasswares with acetone after experiment is completed	**Chemical and fire hazard:** Acetone is an irritant and is flammable	Can cause irritation to skin and eyes. Flammable substance that can start a fire.	1. Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment 3. Avoid placing acetone near any open or naked flame 4. Dispose the washing solvent (acetone) and chemical used in each apparatus into the appropriate halogenated, non-halogenated or heavy metal waste solvent bottle 5. In case of fire, use dry sand, dry chemical, alcohol resistant foam, water spray, or carbon dioxide for extinction	2	1	2			
		Physical hazard: Broken Glassware	Cuts and injury from broken glassware	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 2. Handle the glasswares with care to avoid dropping them. 3. Dispose the broken glassware into the appropriate Sharps bin.	2	1	2			

Conducted By _____

Name Choi Seung Eun

Signature Choi Seung Eun

date 19/9/2024

Approved By

Name Dr. Edith Chan

Signature _____

Approval date _____

Next Revision date _____
(Maximum 3 years)

Risk Assessment Guide

		Likelihood		
		Likely	Possibly	Unlikely
Severity	Low	3	2	1
	Med	6	4	2
	High	9	6	3

** **Risk = Likelihood x Severity**

RISK	DECISION PROCESS
< 3	**RISK ACCEPTABLE**
3, 4	**CONSIDER ADDITIONAL RISK CONTROL**
> 4	**ADDITIONAL RISK CONTROL REQUIRED**

Likelihood

1	**Unlikely**	Not likely to occur (has not occurred in the PI's Lab or similar Lab setup.)
2	**Possible**	Possible or known to occur (has occurred in the PI's Lab or Similar Lab setup.)
3	**Very Likely**	Common or repeating occurrence (has occurred repetitively in the PI's Lab or similar Lab setup.)

Severity

1	**Low**	(e.g. No injury, injury or ill-health requiring first aid treatment only - includes minor cuts and bruises, irritation, ill-health with temporary discomfort)
2	**Medium**	(e.g. Injury requiring medical treatment or ill-health leading to disability – includes lacerations, burns, sprains, minor fractures, dermatitis, deafness, work-related upper limb disorders)
3	**High**	(e.g. Fatal, serious injury or life-treatening occupational disease – includes amputations, major fractures, multiple injuries, occupational cancer, acute poisoning and fatal diseases)

Severity - Consider the magnitude/severity of the consequences of the Risk Factor occurring and then list this as 3 (High), 2 (Moderate) or 1 (Low).

Likelihood - Team should rely upon their experience and consider realistic scenarios. Listed below are examples of factors that may be considered in determining the likelihood.
- Past experience / incidents
- Complexity of the activity
- Number of personnel involved in the activity (e.g. all personnel, a limited number of trained personnel, etc)
- Frequency of use or execution
- Degree of control (involvement of contractors)
- Strength/completeness of administrative controls
- Sufficiency/formality of training
- Other....